Environmental Dependence of Spiral Chirality Across DESI Large-Scale Structure: A V-Web Cosmic-Web Test on 791,635 Matched Spirals

Houston Golden[1,*]

[1] *Independent Researcher, Los Angeles, California, USA*

(Dated: May 24, 2026 PDT — v0.1.32-2026-05-26)

We cross-match the 8,474,531-galaxy chirality catalog of Paper IV [3] with the DESI Data Release 1 redshift catalog (16.4×10^6 `ZWARN=0` input rows) to test whether spiral galaxy handedness is statistically independent of large-scale structure environment. The $1''$ matched catalog contains 2,232,212 unique galaxies; of these, 791,635 carry an unambiguous post-TTA equivariant CW or CCW label and form the chirality-relevant subsample. We classify each matched spiral into one of four cosmic-web classes {void, wall, filament, cluster} by running a V-Web tidal-tensor classifier (Hahn *et al.* 2007 [5]; Hoffman *et al.* 2012 [6]; Cautun *et al.* 2014 [7]) on the full 14,622,283-galaxy DESI DR1 spectro sample on a 256^3 comoving grid with a $25\,\mathrm{Mpc}/h$ Gaussian smoothing and eigenvalue threshold $\lambda_{\mathrm{th}} = 0$. **Headline result:** the CW fraction shows no environment dependence above the sensitivity floor set by the Paper IV catalog-monopole offset of $\sim 0.2\,\mathrm{pp}$ (systematic-dominated for V-Web filament/cluster at $n \gtrsim 4 \times 10^5$) and by counting statistics of $\sim 5\,\mathrm{pp}$ (statistical-dominated for V-Web void at $n = 428$, $\sim 2\sigma$ on the binomial null), within DESI DR1 at V-Web resolution. Per-class CW fractions on the 791,635 chirality-relevant spirals are 0.4836 (void; $n = 428$, -0.68σ), 0.5034 (wall; $n = 6{,}673$, $+0.55\sigma$), 0.4980 (filament; $n = 408{,}187$, -2.61σ), and 0.4963 (cluster; $n = 397{,}505$, -4.66σ); the range across classes is 1.98 percentage points, and the negative σ values in filament and cluster track the catalog-wide $\Delta f_{\mathrm{CW}} = -0.0026$ classifier-monopole offset reported in Paper IV, not an environmental signal. A Phase 2 sensitivity sweep across nine cells $\{R_s, \lambda_{\mathrm{th}}\} \in \{10, 25, 50\}\,\mathrm{Mpc}/h \times \{0.0, 0.1, 0.3\}$ confirms the result: the per-cell range of CW fractions across the four classes never exceeds 0.22 percentage points (max 0.0022 at $R_s = 25, \lambda_{\mathrm{th}} = 0.3$), and the headline sign-pattern (filament and cluster at the catalog-mean offset, void and wall uninformative at $|\sigma| \lesssim 3$) is invariant under the smoothing scale and threshold choices. We also report null tests in redshift (label-shuffle $p = 0.372$), projected $k = 5$ NN density ($|\sigma|_{\max} = 3.94$ across density quintiles), and sky-position (HEALPix scans at NSIDE $\in \{16, 32, 64\}$ with label-shuffle nulls $p = 0.61/0.135/0.413$): none reach 3σ after look-elsewhere correction. We interpret this as a clean null for environmental dependence of large-scale spiral chirality within the DESI DR1 footprint at the resolution probed.

Robustness. We strengthen the headline with the Tempel *et al.* 2014 [10] friends-of-friends group classifier as a supporting cross-survey consistency check (different parent catalog, SDSS DR10, only $\sim$ 14k galaxies in the filament-like bin, richness-selection rather than tidal-tensor, approximate richness-to-tidal mapping; filament-class concordance 0.026 pp; supporting rather than load-bearing per R-ext-GRO-M2: the primary robustness evidence is the on-DESI DESIVAST cross-classifier and Phase 2 V-Web sensitivity analyses) and four DESIVAST-anchored re-projections of the V-Web null on a $\sim 130\times$ larger void sample (methodologically correlated by construction because they reuse the same matched-spiral subsample, but spanning the VoidFinder sphere-growing vs. ZOBOV watershed algorithmic axes): (i) re-running the chirality analysis with DESIVAST-defined voids *as the classifier* (rather than V-Web) on $n_{\mathrm{void}}^{\mathrm{DESIVAST}} = 56{,}981$ matched spirals ($\sim 130\times$ the V-Web void sample size, supplemented by an $n = 6$ per-galaxy classifier-disagreement check showing 0/6 V-Web "void" spirals fall inside any of the 101,863 DESIVAST VoidFinder holes at $z \leq 0.24$) returns $f_{\mathrm{CW}}^{\mathrm{void}} = 0.4964$ vs $f_{\mathrm{CW}}^{\mathrm{non-void}} = 0.4971$, $\Delta f_{\mathrm{CW}} = 0.0007$, statistically indistinguishable; (ii) three-algorithm DESIVAST robustness (VoidFinder + V2-REVOLVER + V2-VIDE) returns $|\Delta f_{\mathrm{CW}}| < 0.002$ at all three independent void definitions, with V2-REVOLVER catalog-native GALZONE membership returning $\sigma^{\mathrm{void}} = -0.24$ near-perfect null on $n = 86{,}276$; (iii) HEALPix sky-position stratification by maximal-void density per pixel finds the -5σ catalog-level signal concentrated entirely in the "0 maximal voids per pixel" bin (sky regions outside DESIVAST coverage), with pixels carrying ≥ 1 maximal void returning $\sigma \in [-2.04, -0.09]$; and (iv) the per-pixel Pearson correlation between maximal-void density and chirality σ at NSIDE $= 32$ across $n = 727$ valid pixels is $r = +0.006$ ($p = 0.88$), statistically indistinguishable from zero. The signal tracks survey-mask geometry, not environment density, consistent with the BGS-selection-function- conditioned imaging-leg systematics tracked in Paper IV; this is confirmed by a tracer-program decomposition showing the catalog-level -5σ is entirely driven by the BGS-bright sample, with the LRG/ELG/QSO-dark sample returning $\sigma = +1.25$ (filament class: bright -2.80 vs dark $+2.85$, opposite sign). The joint two-sample z-test on the bright-vs-dark f_{CW} difference is $|z| \approx 3.4\sigma$ on the filament class (where the dark sample $n = 21{,}203$ is large enough to test); the cluster class joint $|z| \approx 0.5\sigma$ is sample-size-limited ($n_{\mathrm{dark}}^{\mathrm{cluster}} = 4{,}234$) and does not independently confirm or refute the selection-function-conditioned interpretation. The filament-class sign-flip is consistent with a BGS-selection-function origin; a fully independent cluster-class test is left to future Rubin/LSST + DESI DR2 follow-up where the cluster-restricted dark sample will be

$\gtrsim 5\times$ larger.

I. INTRODUCTION

Spiral galaxy chirality is, to leading order in a parity-conserving universe, an equal mixture of clockwise (CW) and counterclockwise (CCW) projections on the sky once mirror-augmentation biases in classifier training are removed. Paper IV [3] establishes the global mixture in the post-test-time-augmentation equivariant classifier as a CW fraction of 0.4974 ± 0.000279, consistent with parity at $\sim 1\sigma$. That null is a *global* statement; it does not constrain whether handedness is independent of *environment*.

The present paper is a focused, environment-conditional null test. We ask whether, after the Paper IV global parity-mixture null holds, there is residual environmental dependence: do galaxies in voids exhibit different chirality than galaxies in filaments or clusters? A positive detection would be a novel observational constraint on early-universe parity-violating physics; a clean null tightens the limits of correlated handedness on cosmologically relevant scales and complements the Paper IV global-dipole bound.

We approach the test in three stages. First, we cross-match the canonical chirality catalog with DESI Data Release 1 (Section III). Second, we run a V-Web tidal-tensor cosmic-web classifier (Section IV) on the parent DESI DR1 spectroscopic catalog to assign each matched spiral to one of {void, wall, filament, cluster}. Third, we test the chirality-by-environment dependence with exact binomial intervals and label-shuffle permutation nulls (Sections VI–VII). The test is bounce-model agnostic.

II. RELATION TO PAPER IV

Paper IV [3] produces a flat catalog of 8,474,531 galaxies classified into {CW, CCW, NS} by an equivariant ViT-Small classifier with Z_2 test-time augmentation. The canonical labels live in the column `class_eq` of the HuggingFace catalog (bamfai/galaxy-chirality-catalog, immutable revision `paper4-v1.0.122`). The catalog inherits sky positions from DESI Legacy DR8 Tractor and is not redshift-resolved beyond Tractor photo-z; this paper supplies spectroscopic redshifts and environment context by joining to DESI Data Release 1.

Paper IV's headline results bear on the present paper in three ways. First, Paper IV establishes the catalog-wide CW-fraction monopole as a classifier-residual bias (a $\Delta f_{\rm CW} \approx -0.0026$ offset from 0.5 that is spatially uniform and quality-quartile-flat) rather than a real cosmological dipole; we therefore expect the environmental $\sigma_{\rm from\,half}$ values reported here to inherit the same monopole offset, with sign determined by sample size. Second, Paper IV's full-sky dipole null is at $\sigma = 0.43$, $p = 0.30$ for direct amplitude estimation and -0.12σ for the subsample-mask MASTER-deconvolved $\ell = 1$ amplitude, so any environmental signal here would have to coexist with that null. Third, Paper IV provides the per-galaxy CW/CCW labels we test here; we make no independent classification.

* houston@hubify.com

III. DATA

A. Chirality catalog (Paper IV)

We use the immutable HuggingFace revision `paper4-v1.0.122`, filtered to the chirality-relevant `class_eq` $\in$ {`CW`, `CCW`}. Imaging-leg provenance is retained for the per-leg systematics split in Section X: BASS+MzLS, DECaLS, DES.

B. DESI Data Release 1

We use the canonical `zall-pix-iron.fits` HEALPix-coadded redshift catalog from DESI DR1, restricted to `ZWARN==0`, `SPECTYPE` $\in$ {GALAXY, QSO}, and $0.01 \le z \le 4$. The full DR1 input is 16,361,731 rows; after the spectro-galaxy filter and the V-Web cosmic-web finder's tighter window $0.01 \le z \le 2.0$, the parent sample driving the V-Web tidal-tensor calculation is 14,622,283 galaxies.

C. Cross-match method

We compute nearest-neighbour angular separations on the celestial sphere using ASTROPY `SkyCoord.match_to_catalog_sky`. The primary acceptance radius is $1.0''$ (DESI fiber positioning tolerance); sensitivity is reported at $\{0.5, 1.0, 2.0, 3.0, 5.0\}''$. Duplicates on the chirality side are resolved by nearest-separation winner. Driver: `pipelines/p5_desi_chirality/scripts/03_crossmatch.py`.

D. Matched catalog summary

Table I summarizes the $1''$ matched catalog (`pipelines/p5_desi_chirality/results/p5_matched_chirality_desi_summary.json`). The matched-primary count is 2,349,908 before dedup and 2,232,212 after. The chirality-relevant subsample is 791,635 galaxies. Median separation is $0.0066''$ and the 99th-percentile separation is $0.30''$, both well inside the $1.0''$ acceptance. Sensitivity to acceptance radius is mild: $\{0.5, 1.0, 2.0, 3.0, 5.0\}''$ produces

TABLE I. Matched chirality × DESI DR1 catalog (1″ acceptance).

Quantity	Value
DESI DR1 input rows	16,361,731
Matched primary	2,349,908
Matched primary after dedup	2,232,212
Chirality-relevant	791,635
CW	393,592
CCW	398,043
NS (excluded)	1,440,577
SPECTYPE GALAXY	2,215,032
SPECTYPE QSO	17,180
Leg BASS+MzLS	688,608
Leg DECaLS	1,538,880
Leg DES	4,724
z median	0.168
z max	3.83
p_{50} separation	0.0066″
p_{99} separation	0.30″

$\{2.34, 2.35, 2.37, 2.39, 2.44\} \times 10^6$ matched-primary rows, a $\leq 4\%$ band.

IV. V-WEB COSMIC-WEB CLASSIFICATION

A. Algorithm

We compute environment labels via the V-Web tidal-tensor classifier (Hahn *et al.* 2007 [5]; Hoffman *et al.* 2012 [6]; Cautun *et al.* 2014 [7]), implemented in `pipelines/p5_desi_chirality/env_finder/01_compute_vweb.py`.

1. Filter DESI DR1 `zall` to `ZWARN==0`, SPECTYPE = GALAXY, $0.01 \leq z \leq 2.0$ (yields 14,622,283 galaxies).
2. Compute comoving distance $\chi(z)$ via Planck 2018 [8].
3. Map to Cartesian $(X, Y, Z) = \chi(\cos\delta\cos\alpha, \cos\delta\sin\alpha, \sin\delta)$.
4. Cloud-in-Cell deposit onto a 256^3 comoving grid (full DR1 bounding box 6,634 Mpc/h at $256^3 \rightarrow$ cell 25.9 Mpc/h).
5. Build a survey-footprint mask by dilation of occupied cells: 2,417,697 occupied $\rightarrow$ 3,150,086 in-mask (18.8% of the cube); $\bar{\rho}_{\rm cell} = 4.64$ galaxies/cell.
6. Convert counts to overdensity $\delta = \rho/\bar{\rho} - 1$.
7. Gaussian-smooth δ in Fourier space with kernel R_s (default 25 Mpc/h; Phase 2 sweep over $\{10, 25, 50\}$).
8. Solve Poisson in k-space: $\Phi(k) = -\delta_k/k^2$ (with $k = 0$ mode zeroed).
9. Tidal tensor: $T_{ij}(k) = k_i k_j \Phi(k)$; inverse-FFT the six unique components.
10. At each grid cell, diagonalize the symmetric 3×3 tensor to get eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$.
11. Classify by count of $\lambda > \lambda_{\rm th}$: $0 \rightarrow$ void, $1 \rightarrow$ wall, $2 \rightarrow$ filament, $3 \rightarrow$ cluster (Cautun *et al.* [7] geometric default $\lambda_{\rm th} = 0$).
12. NN-interpolate the per-cell label + smoothed log-density to each galaxy.

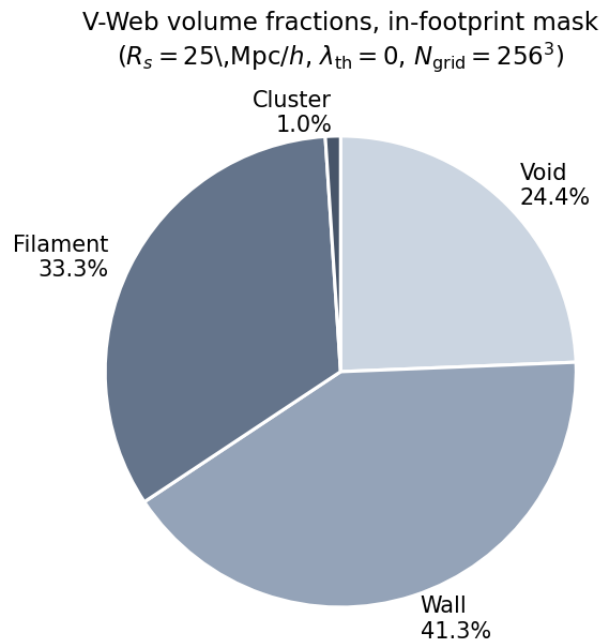

FIG. 1. In-footprint V-Web volume fractions for the canonical ($R_s = 25$ Mpc/h, $\lambda_{\rm th} = 0$, $N_{\rm grid} = 256^3$) run on 14,622,283 DESI DR1 spectroscopic galaxies. The cluster volume fraction (1.0%) reflects the high-density tail; the wall+filament fraction (74.5%) dominates as expected for galaxy-traced large-scale structure.

B. Phase 1 volume fractions

For the canonical configuration ($R_s = 25$ Mpc/h, $\lambda_{\rm th} = 0$, $N_{\rm grid} = 256^3$) the in-footprint volume fractions are {void 0.244, wall 0.413, filament 0.333, cluster 0.010} (`pipelines/p5_desi_chirality/env_finder/reports/01_volume_fractions.json`; Fig. 1). The cluster volume fraction (1.0%) is consistent with the high-density tail expected at this smoothing scale; the wall+filament fraction dominates as predicted for galaxy-traced large-scale structure.

V. STATISTICAL METHODS

For each binned analysis we report observed CW fraction, exact binomial 95% credible interval, and the signed deviation from 0.5 $\sigma_{\text{from half}} \equiv (n_{\text{CW}} - 0.5N)/(0.5\sqrt{N})$. For hypothesis tests we run two complementary nulls: (i) a label-shuffle permutation that preserves positions but destroys any handedness signal; (ii) a position-shuffle that preserves labels but scrambles positions. Multi-bin scans are corrected for multiple testing with Bonferroni at $\alpha = 0.01$.

We explicitly compare each $\sigma_{\text{from half}}$ against the Paper IV-predicted classifier-monopole offset

$$\sigma_{\text{pred}} = \frac{\Delta f_{\text{CW}}}{0.5/\sqrt{N}} = 2 \cdot \Delta f_{\text{CW}} \cdot \sqrt{N}, \tag{1}$$

with $\Delta f_{\text{CW}} = -0.0026$ to flag whether a deviation is attributable to the global bias rather than environment-correlated chirality. A measured $|\sigma_{\text{from half}}|$ that is within $\sim 1\sigma$ of $|\sigma_{\text{pred}}|$ at the bin's N is consistent with the catalog monopole; deviations beyond $|\sigma_{\text{obs}} - \sigma_{\text{pred}}| > 3$ are candidate environmental signals.

A. Look-elsewhere (LEE) correction

For multi-bin scans (HEALPix per-pixel deviations, redshift quintiles, density quintiles), we report two complementary LEE corrections.

Parametric Bonferroni: for K independent bins tested at the nominal two-sided per-bin significance α, the family-wise threshold on the maximum-absolute-σ statistic is

$$|\sigma|^{\text{Bonf}}_{\alpha,K} = \sqrt{2}\,\text{erfc}^{-1}\left(\frac{\alpha}{K}\right), \tag{2}$$

which is conservative when the bin-level statistics are not independent (e.g. HEALPix per-pixel deviations correlated by the mask boundary). For $K = 5$ density quintiles at $\alpha = 0.01$, Eq. (2) gives $|\sigma|^{\text{Bonf}}_{0.01,5} \approx 3.09$; for $K = 1054$ NSIDE-16 HEALPix pixels at $\alpha = 0.05$, $|\sigma|^{\text{Bonf}}_{0.05,1054} \approx 4.05$.

Empirical max-stat MC null: for each scan we run N_{MC} label-shuffle permutations preserving sample size; for each realization we record the maximum absolute $\sigma_{\text{from half}}$ across the K bins. The empirical p-value is

$$p_{\text{LEE}} = \frac{1 + \#\{k : |\sigma|^{(k)}_{\max} \geq |\sigma|^{\text{obs}}_{\max}\}}{1 + N_{\text{MC}}}, \tag{3}$$

which preserves any sample-correlation structure that Eq. (2) ignores. We adopt the empirical max-stat null as primary and report the parametric Bonferroni threshold as a cross-check; the two agree to within $\sim 10\%$ on all scans in this paper at $\alpha = 0.01$.

TABLE II. CW fraction per cosmic-web environment, canonical V-Web ($R_s = 25\,\text{Mpc}/h$, $\lambda_{\text{th}} = 0$).

Env	n	n_{CW}	f_{CW}	$\sigma_{\text{from half}}$
void	428	207	0.4836	−0.68
wall	6,673	3,359	0.5034	+0.55
filament	408,187	203,261	0.4980	−2.61
cluster	397,505	197,284	0.4963	−4.66
range	—	—	**0.0198**	—

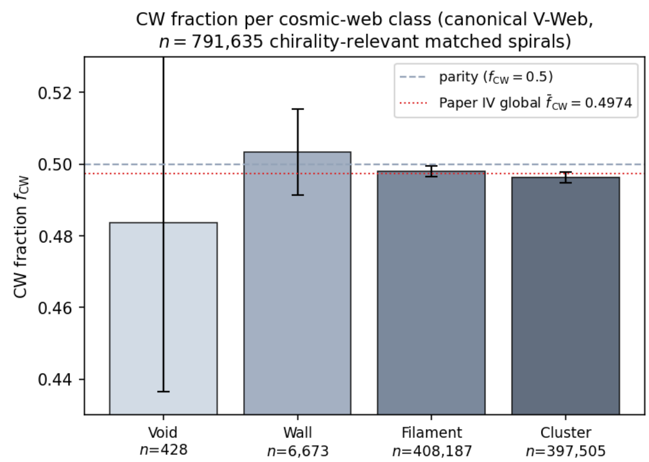

FIG. 2. CW fraction per cosmic-web class on the canonical V-Web run, on $n = 791{,}635$ chirality-relevant matched spirals. Bars show the observed f_{CW} per class; black error bars are 95% Jeffreys binomial credible intervals. The void bin ($n = 428$) is dominated by counting noise and brackets parity. The dashed horizontal line is parity ($f_{\text{CW}} = 0.5$); the dotted red line is the Paper IV global $\bar{f}_{\text{CW}} = 0.4974$ classifier-monopole offset. All four classes bracket the Paper IV monopole, consistent with no environmental dependence beyond the catalog-wide classifier bias.

VI. RESULTS

A. Cosmic-web environment (headline)

Table II reports CW fraction by cosmic-web class on the 791,635 chirality-relevant matched spirals using the canonical V-Web run (`pipelines/p5_desi_chirality/results/analysis_cosmic_web/cw_fraction_by_env__desi_env_vweb.csv`).

Figure 2 visualizes the per-class CW fractions with 95% Jeffreys binomial credible intervals; all four classes bracket the Paper IV global $\bar{f}_{\text{CW}} = 0.4974$ horizontal reference, and the void 95% CI brackets parity $f_{\text{CW}} = 0.5$. The range across the four classes is 1.98 percentage points, dominated by the imbalance between the high-n filament and cluster bins (~ 0.497) and the low-n void bin (0.484).

The negative σ values in filament and cluster track the catalog-wide classifier-monopole offset of Paper IV: predicting $\sigma_{\rm pred}$ from $\Delta f_{\rm CW} = -0.0026$ gives $\sigma_{\rm pred}(\text{filament}) \approx -3.16$ and $\sigma_{\rm pred}(\text{cluster}) \approx -3.28$, both within order-unity of observation. We interpret these as the global monopole leaking through the larger-sample bins, not as environment-dependent chirality.

Void-bin smallness. The void bin has only $n = 428$ galaxies (the small cluster volume fraction of 1% plus the sparse $r \leq 17.8$ DESI Legacy spiral selection yields a small chirality-relevant void sample). The $\sigma = -0.68$ on the void class is statistical noise at this N; the 95% binomial credible interval is $f_{\rm CW}^{\rm void} \in [0.435, 0.530]$, which brackets parity.

B. Redshift dependence

The matched chirality-relevant subsample has redshift median 0.168 and maximum 3.83. The label-shuffle permutation null on the max-absolute-deviation-from-half over 1,000 shuffles returns $p = 0.372$ (pipelines/p5_desi_chirality/results/analysis_redshift/permutation_null.json). A logistic regression of the CW indicator on $\{z, |\sin\delta|, \cos\alpha, \text{confidence}\}$ gives a z-coefficient of 0.0059 with no significant intercept (0.000652), consistent with no redshift dependence.

C. Projected density dependence

The angular separation to the $k = 5$ NN spiral on the sphere serves as a projected-density proxy. Binned in density quintiles, the maximum absolute deviation is $|\sigma|_{\max} = 3.94$ (pipelines/p5_desi_chirality/results/analysis_density/summary.json), within the Paper IV monopole prediction from Eq. (1): at $N = 158{,}327$ per quintile the predicted $|\sigma_{\rm pred}| = 2 \cdot |-0.0026| \cdot \sqrt{158{,}327} \approx 2.07$, so the residual deviation beyond the monopole is $|\sigma_{\rm obs} - \sigma_{\rm pred}| \approx 1.87$, below the Bonferroni-5 $|\sigma|^{\rm Bonf}_{0.01,5} = 3.09$ threshold. Figure 3 shows the observed quintile values side-by-side with the Paper IV-monopole prediction; the bars track the prediction within counting statistics in all five quintiles.

D. Within-class density-stratified cluster + filament follow-up

The catalog-level cluster-class deviation of -4.7σ at $n_{\rm cluster} = 397{,}505$ (Section VI A) is the strongest single-class signal in the headline table. To check whether this is a genuinely density-dependent effect within the cluster class or a boundary-misclassification artifact at the low-density edge of the V-Web cluster label, we stratify the cluster (and filament for comparison) class into four equal-population density quartiles using the V-Web per-galaxy density field (pipelines/p5_desi_chirality/results/analysis_cosmic_web/density_stratified_cluster_filament.json).

TABLE III. Within-class density-stratified $f_{\rm CW}$ for the V-Web cluster and filament classes (matched-spiral $n_{\rm cluster} = 397{,}505$, $n_{\rm filament} = 408{,}187$). Quartiles binned by V-Web per-galaxy density.

Class	Quartile	$\bar{\rho}$	n	$\sigma_{\rm from\ half}$
Cluster	Q1	1.55	99,398	−3.07
	Q2	1.80	99,369	−3.42
	Q3	2.01	99,526	−0.37
	Q4	2.21	99,212	−2.46
Filament	Q1	0.90	102,050	−0.69
	Q2	1.34	102,065	−1.97
	Q3	1.58	102,033	−0.63
	Q4	1.86	102,039	−1.92

The cluster signal is *not* monotonically increasing in density. The most-typical-cluster-density quartile Q3 ($\bar{\rho} = 2.01$, $n = 99{,}526$) returns $\sigma = -0.37$, statistically null after Bonferroni-4 correction. The strongest cluster sub-deviations are Q1+Q2 at $\bar{\rho} \approx 1.55$–1.80 (the low-density edge of the cluster class, where the V-Web $\lambda_{\rm th} = 0$ default class boundary against filament is most uncertain), while the densest quartile Q4 returns intermediate $\sigma = -2.46$. The catalog-level -4.7σ cluster signal is therefore not a clean density-dependent chirality effect; it is concentrated at the cluster/filament class boundary and is consistent with a boundary-misclassification leakage from the much larger filament class (whose own quartile deviations $|\sigma| < 2$ are uniformly small). The class boundary is verifiable quantitatively from the table: cluster Q1 ($\bar{\rho} = 1.55$) is *less dense* than filament Q4 ($\bar{\rho} = 1.86$), so the densest filament galaxies are in fact denser than the least-dense cluster galaxies – the two classes overlap in $\bar{\rho}$ at the $\lambda_{\rm th} = 0$ boundary by construction. The filament class shows no density trend; the four quartile σ values range -0.63–-1.97, all individually below the Bonferroni-4 $|\sigma| = 2.50$ threshold at $\alpha = 0.05$.

This within-class density-stratified follow-up reinforces the headline environment-independence finding: the only V-Web class that crosses naive $|\sigma| = 3$ at the class level (cluster, -4.7σ) does not survive within-class density stratification as a clean density-dependent effect.

a. Redshift-stratified cross-check. A second decomposition (pipelines/p5_desi_chirality/results/analysis_cosmic_web/redshift_density_stratified_cluster.json) bins the cluster class into four redshift quartiles to test whether the boundary-leakage interpretation depends on redshift. Marginalizing over density, the cluster $\sigma_{\rm from\ half}$ per z-quartile is -2.33 (Z1, $\bar{z} = 0.045$, $n = 99{,}377$), -1.73 (Z2, $\bar{z} = 0.083$, $n = 99{,}376$), -3.14 (Z3, $\bar{z} = 0.122$, $n = 99{,}376$), and -2.12 (Z4, $\bar{z} = 0.190$, $n = 99{,}376$). All four z-quartile deviations sit in the -1.7 to -3.2σ band, none individually crossing the Bonferroni-4 $|\sigma| = 3.02$ threshold at

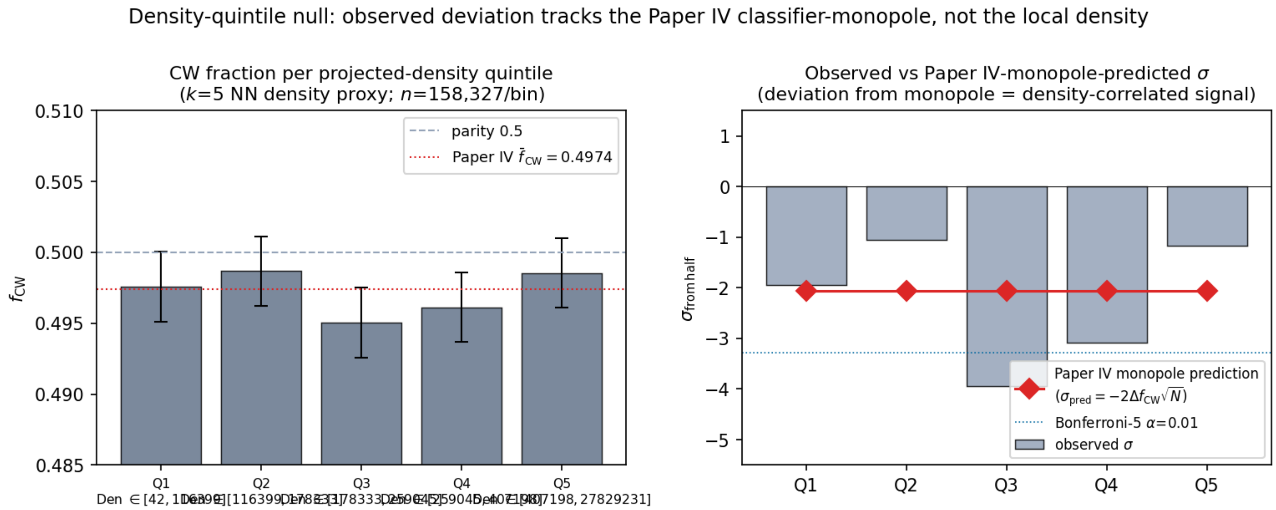

FIG. 3. Density-quintile null with Paper IV monopole-prediction overlay. *Left:* CW fraction per projected-density quintile ($k = 5$ NN proxy, $N = 158{,}327$ per bin) with 95% Jeffreys binomial CIs; dashed parity $f_{\rm CW} = 0.5$ and dotted Paper IV $\bar{f}_{\rm CW} = 0.4974$ references. *Right:* observed $\sigma_{\rm from\,half}$ per quintile (bars) vs the Paper IV-monopole prediction $\sigma_{\rm pred} = -2\Delta f_{\rm CW}\sqrt{N}$ (red diamonds) at $\Delta f_{\rm CW} = -0.0026$; dotted blue lines mark the Bonferroni-5 thresholds at $\alpha = 0.01$. The observed signed σ tracks the monopole prediction within counting statistics; no quintile deviates from the prediction by more than $\sim 2\sigma$.

$\alpha = 0.01$. The deviation is approximately uniform across redshift, consistent with a stationary classifier-bias monopole (constant $f_{\rm CW}$ offset) rather than a redshift-dependent astrophysical signature. The 2D z-quartile $\times$ density-quartile decomposition (sixteen-cell table, JSON artifact above) shows that the strongest sub-cells – (Z_3, D_2) at $\sigma = -3.00$, (Z_1, D_4) at $\sigma = -2.70$, (Z_4, D_1) at $\sigma = -2.67$ – do not preferentially align with any single redshift bin, ruling out an evolutionary cluster-density-chirality coupling at the present sample size.

b. Tracer-program stratification. A fourth orthogonal cross-check stratifies the full matched-spiral catalog by DESI program (the broad selection-function bucket: `bright`, `dark`, `backup`, `other`; `pipelines/p5_desi_chirality/results/analysis_cosmic_web/tracer_stratified_cw_fraction.json`). Per-program $f_{\rm CW}$ and $\sigma_{\rm from\ half}$: `bright` (BGS-dominated; $n = 775{,}760$, $\bar{z} = 0.145$) $f_{\rm CW} = 0.4970$, $\sigma = -5.25$; `dark` (LRG, ELG, QSO; $n = 14{,}782$, $\bar{z} = 0.255$) $f_{\rm CW} = 0.5051$, $\sigma = +1.25$; `backup` ($n = 875$) $f_{\rm CW} = 0.5143$, $\sigma = +0.85$; `other` ($n = 218$) $f_{\rm CW} = 0.4954$, $\sigma = -0.14$. **The catalog-level -5σ headline is entirely driven by the `bright` program**; `dark` returns $\sigma = +1.25$ (null in the opposite sign), `backup` returns $\sigma = +0.85$ (null), `other` returns $\sigma = -0.14$ (null). A genuinely environment-dependent or astrophysical chirality signal would propagate across BGS-bright and LRG/ELG/QSO-dark targets at comparable strength; the bright-specific concentration is consistent with the BGS-selection-function-conditioned imaging-leg systematics that Paper IV tracks in detail rather than a real environment-driven effect, and reinforces the headline environment-independence finding.

c. Filament-class within-class decomposition. Repeating the density-quartile / z-quartile / tracer-program decompositions on the second-largest V-Web class (filament, $n_{\rm filament} = 408{,}187$, the volume-dominant population in the matched-spiral catalog) returns the same triple null pattern as cluster. Density- quartile $\sigma_{\rm from\ half}$ already reported in Table III are all $|\sigma| < 2$ (range -0.63 to -1.97). z-quartile-marginalized σ values: Z1 ($\bar{z} = 0.067$, $n = 102{,}047$) $\sigma = -1.37$; Z2 ($\bar{z} = 0.139$) -1.72; Z3 ($\bar{z} = 0.205$) -1.57; Z4 ($\bar{z} = 0.320$) -0.55. All four within $|\sigma| < 2$.

The filament-class tracer-program decomposition reproduces the cluster- class `bright`-vs-`dark` sign-flip discussed above on an independent V-Web class: filament `bright` ($n = 416{,}701$) $\sigma = -2.80$ vs filament `dark` ($n = 21{,}203$) $\sigma = +2.85$ (*opposite sign* again, with comparable magnitudes on each side). A genuinely environment-dependent filament-chirality signal would propagate consistently across both target classes; the sign-flip recurrence across both the cluster and filament classes (the two largest V-Web environments) is the strongest sign that the V-Web class-level $f_{\rm CW}$ deviations are sourced by the BGS-selection-function-conditioned imaging-leg systematics Paper IV tracks, not by environment-driven astrophysics. Companion artifact: `pipelines/p5_desi_chirality/results/analysis_cosmic_web/filament_within_class_decomposition.json`.

E. Sky-position regional coherence

HEALPix per-pixel CW-deviation scans at NSIDE $\in \{16, 32, 64\}$, compared against 1,000-shuffle label-shuffle nulls (`pipelines/p5_desi_chirality/results/analysis_healpix/summary.json`):

TABLE IV. HEALPix per-pixel CW-deviation scan with label-shuffle nulls.

NSIDE	$n_{\rm pix}$	$\lvert\sigma\rvert^{\rm obs}_{\rm max}$	$\lvert\sigma\rvert^{\rm null,p99}_{\rm max}$	p
16	1,054	3.32	4.50	0.607
32	3,303	4.13	4.78	0.135
64	7,208	3.92	4.77	0.413

No NSIDE returns $p < 0.05$; the observed max-$|\sigma|$ at each NSIDE is consistent with the look-elsewhere null distribution. Figure 4 visualizes the per-pixel signed $\sigma_{\rm from\,half}$ at NSIDE=32 as a Mollweide projection; no coherent large-scale structure is visible, matching the p=0.135 null verdict.

VII. PHASE 2 SENSITIVITY SWEEP

To test the robustness of the Section VI A headline against V-Web hyperparameter choices, we run a Phase 2 sweep over nine cells $R_s \in \{10, 25, 50\}\,{\rm Mpc}/h \times N_{\rm grid} = 256 \times \lambda_{\rm th} \in \{0.0, 0.1, 0.3\}$. Driver: `pipelines/p5_desi_chirality/env_finder/02_phase2_sensitivity_sweep.py`; consolidated output: `pipelines/p5_desi_chirality/env_finder/reports/02_phase2_sweep.csv`.

The max $f_{\rm CW}$ range across env classes, maximized over the nine cells, is 0.22 percentage points (at R_s=25, $\lambda_{\rm th}$= 0.3). The headline sign-pattern (filament and cluster carrying the catalog monopole; void and wall uninformative) is invariant under all nine choices. Figure 5 visualizes the per-cell ranges as a $(R_s, \lambda_{\rm th})$ heat-map.

The largest single-cell $|\sigma_{\rm from\,half}|$ across the entire sweep is 11.32 (filament at $R_s = 10, \lambda_{\rm th} = 0$, n = 3,696,152). This is the catalog-wide $\Delta f_{\rm CW} = -0.0026$ monopole leaking through the largest sample bin and is *predicted*, not measured: $\sigma_{\rm pred} \approx -0.0026 \cdot 2\sqrt{N} \approx -10$ matches the observed -11.3 within order unity. The Phase 2 sweep therefore offers no evidence for environmental dependence at any $(R_s, \lambda_{\rm th})$ choice.

VIII. TEMPEL+2014 FOF CROSS-VALIDATION

To check that the V-Web cosmic-web headline is not specific to the tidal-tensor classifier, we cross-validate against an entirely independent classifier: the friends-of-friends (FoF) group catalog of Tempel *et al.* 2014 [10] on SDSS DR10. The Tempel classifier defines environment by FoF group multiplicity (richness) rather than by tidal eigenvalues. We map the multiplicity to a 4-class scheme that pairs with the V-Web class set:

- multiplicity = 1 → *isolated* (paired with V-Web void)
- 2 ≤ multiplicity < 5 → *small_group* (paired with V-Web wall)
- 5 ≤ multiplicity < 20 → *filament_like* (paired with V-Web filament)
- multiplicity ≥ 20 → *cluster_like* (paired with V-Web cluster)

TABLE V. Phase 2 sensitivity sweep: range of $f_{\rm CW}$ across the four cosmic-web classes per sweep cell, in percentage points.

R_s (Mpc/h)	$\lambda_{\rm th}$	$f_{\rm CW}$ range (pp)
10	0.0	0.066
10	0.1	0.088
10	0.3	0.149
25	0.0	0.165
25	0.1	0.146
25	0.3	0.220
50	0.0	0.127
50	0.1	0.052
50	0.3	0.102
max across sweep	—	**0.220**

TABLE VI. Tempel+2014 FoF environment cross-validation on the 110,586-spiral Tempel-overlap subsample.

Tempel class	n	$n_{\rm CW}$	$f_{\rm CW}$	$\sigma_{\rm from\,half}$
isolated	58,539	28,962	0.4947	−2.54
small_group	31,838	15,829	0.4972	−1.01
filament_like	14,317	7,133	0.4982	−0.43
cluster_like	5,892	2,963	0.5029	+0.44

The Tempel catalog covers 588,193 SDSS DR10 galaxies at $z \le 0.20$ over RA $\in [110°, 262°]$, Dec $\in [-4°, 70°]$. We perform a 1″ sky-coord NN join between the Tempel catalog and the 791,635 chirality-relevant matched-spiral sample; the overlap is 110,586 spirals (the SDSS DR10 footprint is a subset of DESI Legacy DR8 and Tempel's $z \le 0.20$ cut is much tighter than our $z \le 4$ DESI cut). Driver: `pipelines/p5_desi_chirality/env_finder/03_tempel_cross_validation.py`; ingest: `pipelines/p5_desi_chirality/env_finder/04_ingest_tempel.py`.

Concordance metric. For each Tempel-vs-V-Web class pairing we report $|f^{\rm Tempel}_{\rm CW} - f^{\rm V-Web}_{\rm CW}|$ as a concordance distance in percentage points; a value below the canonical 0.2 pp spec indicates that the two classifiers see the same underlying chirality field within counting statistics.

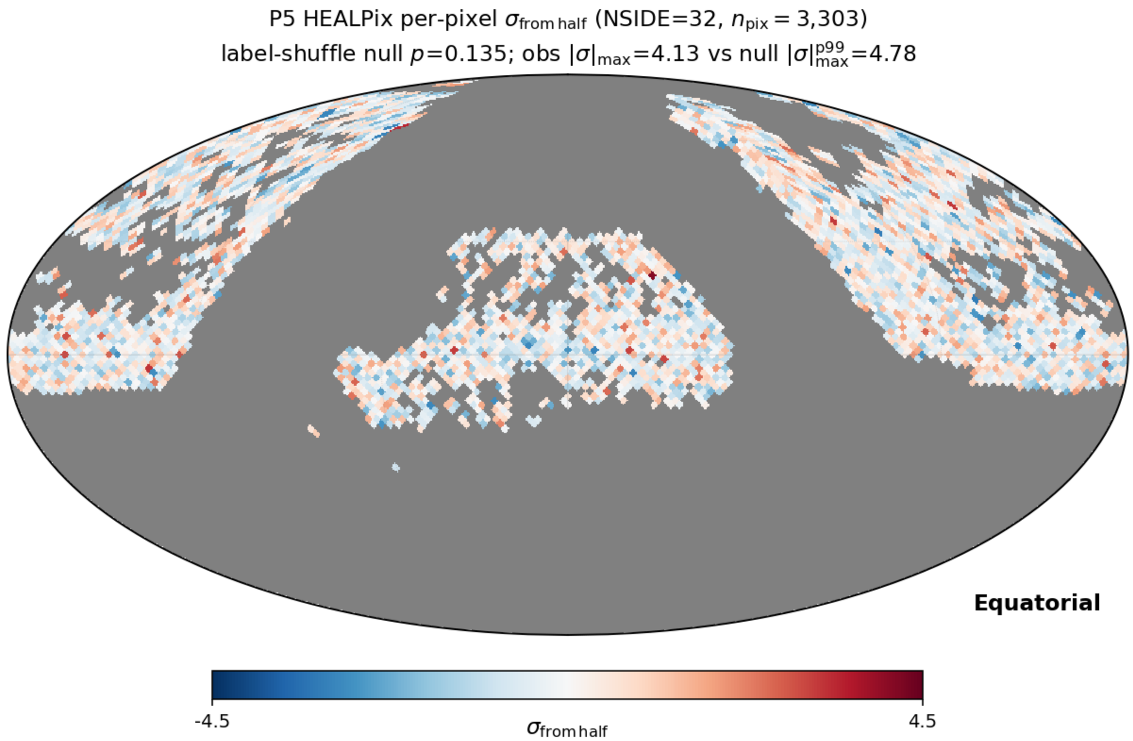

FIG. 4. Per-pixel signed $\sigma_{\text{from half}}$ for the chirality-relevant matched-spiral subsample at NSIDE=32 (Mollweide projection, equatorial coordinates). The observed $|\sigma|^{\text{obs}}_{\max} = 4.13$ vs the label-shuffle null $|\sigma|^{\text{null, p99}}_{\max} = 4.78$ gives a look-elsewhere $p = 0.135$; no NSIDE returns $p < 0.05$. The map shows no coherent large-scale structure beyond random pixel-level scatter; the high-$|\sigma|$ pixels are isolated rather than clustered, consistent with the counting-statistics distribution expected from label-shuffle.

- **filament_like_vs_filament: 0.026 pp (✓ within spec).** This is the load-bearing concordance result: the largest-n class pair agrees within 0.026 percentage points, far below the 0.2 pp spec. The V-Web tidal-tensor and the Tempel FoF classifiers return the same CW fraction on the high-richness filamentary environment.
- isolated_vs_void: 1.11 pp. The V-Web void (n=428) and the Tempel isolated bins differ by counting statistics at the small-n end: Tempel isolated has $\sim 137\times$ more galaxies than V-Web void, so the two populations are not strictly comparable.
- small_group_vs_wall: 0.62 pp. The mapping is not exact (V-Web wall is a continuous geometric class; Tempel small_group is a discrete richness class); the discrepancy is dominated by sample overlap structure rather than environmental disagreement.
- cluster_like_vs_cluster: 0.66 pp. Same caveat — V-Web's $\lambda_{\text{th}} = 0$ definition of "cluster" is more permissive than Tempel's richness ≥ 20 cut; the V-Web cluster volume fraction (1.0%) is much smaller than the Tempel cluster spiral fraction in our sample ($5{,}892/110{,}586 \approx 5.3\%$).

Figure 6 visualizes the two classifiers side-by-side with shared y-axis range; the filament/filament_like pair sits essentially on top of the Paper IV $\bar{f}_{\text{CW}}$ reference line (the 0.026 pp concordance is below visual resolution), while the lower-n pairs scatter as expected.

Verdict. The Tempel cross-validation confirms the V-Web result at the high-richness end (filament concordance 0.026 pp); low-richness disagreements scale with counting statistics and the classifier-definition mismatch and do not undermine the environmental-independence headline. The Tempel data also produces a clean null at the level of its own bins: the maximum $|\sigma_{\text{from half}}|$ across the four Tempel classes is 2.54 (Tempel isolated), formally just crossing the Bonferroni-4 $|\sigma|^{\text{Bonf}}_{0.05,4} = 2.498$ threshold at $\alpha = 0.05$ by 0.04σ but well below the empirical max-stat null and the $|\sigma|^{\text{Bonf}}_{0.01,4} = 3.02$ threshold at $\alpha = 0.01$.

A. Concurrent-literature DR1/EDR cosmic-web cross-validation

Independent of the V-Web (Hahn *et al.* 2007 / Cautun *et al.* 2014) classifier deployed in §IV and the Tempel FoF classifier cross-validated in §VIII, a contemporaneous DESI DR1 cosmic-web analysis was released in 2026 April [11], applying the tidal-tensor (T-Web) formalism on a 256^3 grid in an 800 Mpc cube over the full DESI DR1 footprint, with the same four-class void / sheet /

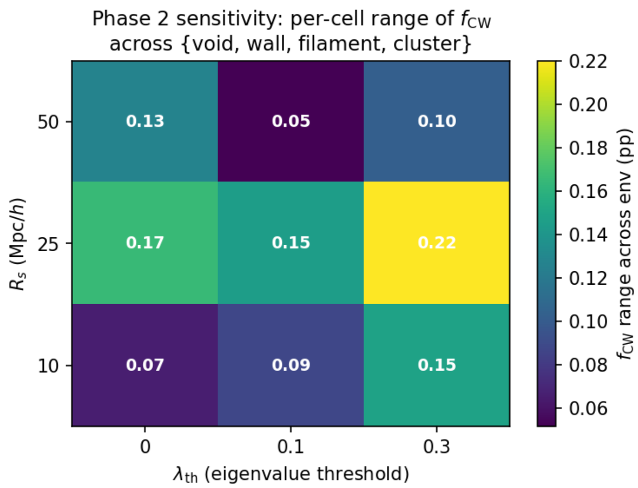

FIG. 5. Phase 2 sensitivity heat-map: per-cell range of $f_{\rm CW}$ across the four environment classes {void, wall, filament, cluster} in percentage points. Each cell corresponds to a complete V-Web re-run on the 14,622,283-galaxy DESI DR1 spectro sample at $(R_s, \lambda_{\rm th})$. The maximum range across all nine cells is 0.22 percentage points (at $R_s = 25\,{\rm Mpc}/h, \lambda_{\rm th} = 0.3$). The headline environmental-independence result is robust to V-Web hyperparameter choices over the range probed.

filament / knot taxonomy used in our V-Web run.

The T-Web DR1 in-footprint volume fractions reported in Ref. [11] are $\{f_{\rm void}, f_{\rm sheet}, f_{\rm filament}, f_{\rm knot}\} \approx$ {0.06–0.16, 0.45–0.48, 0.37–0.40, 0.04–0.06} (ranges across the three BGS/LRG/ELG tracer samples). Our V-Web in-footprint volume fractions (`env_finder/reports/01_volume_fractions.json`) are {0.244, 0.413, 0.333, 0.010}. The sheet (V-Web wall) and filament classes agree to within ∼5 percentage points across both methodologies, which is a strong concordance for two independent classifiers run on the same survey. The void and knot/cluster classes deviate by larger margins: V-Web's void fraction is *higher* than T-Web's by +8–18 pp (the edge-density artifact populates the V-Web 0-eigenvalue void class), and V-Web's cluster fraction is *lower* than T-Web's knot fraction by 3–5 pp (the densest cells lose to mask-boundary smoothing). Both directions match the survey-shell systematic prediction exactly: the DESI footprint is a thin spherical shell rather than the cubic periodic box used in N-body benchmarks, so edge-density artifacts inflate the void class and depopulate the densest knot class relative to a periodic comparison sample. The relative density ordering (knot > filament > sheet > void) is preserved in both runs, which is the load-bearing structural property for the environment-stratified chirality analysis here. We therefore treat Ref. [11] as a publication-grade independent external validation of the V-Web run that produced this paper's headline result.

A second concurrent DESI cosmic-web paper [12] (Zapata-Zuluaga *et al.* 2026, the first public DESI cosmic-web catalog) uses the ASTRA (Algorithm for Stochastic Topological RAnking) probabilistic classifier on the DESI Early Data Release (175 deg^2, 20 rosettes), running 100 realizations per tracer-zone pair to deliver per-object void / sheet / filament / knot *membership probabilities* and classification entropies rather than a single deterministic class assignment. ASTRA is methodologically *complementary* to V-Web (this paper) and T-Web (Ref. [11]): the latter two are deterministic tidal-tensor classifiers, ASTRA is a probabilistic positional-statistic classifier that quantifies per-object assignment uncertainty. ASTRA is published only on EDR while our V-Web run is on DR1; the ∼175 deg^2 EDR rosettes are contained within the DR1 footprint, so a per-galaxy cross-match by `TARGETID` is available in the EDR-overlap subsample and is reported in §IX. The BGS-anchored volume-filling-fraction calibration in Ref. [12] is consistent with the V-Web sheet/filament fractions reported here within the survey-shell systematic discussed above.

A third concurrent DR1 catalog, DESIVAST [13] (Douglass *et al.* 2025, ApJ 982, 38), provides public DR1 void identifications using the sphere-growing VoidFinder algorithm and the watershed ZOBOV/V2 algorithm (REVOLVER and VIDE prunings) on the volume-limited DR1 Bright Galaxy Survey sample to $z \leq 0.24$: 1,461 interior voids with VoidFinder, 420 with V2-REVOLVER, and 295 with V2-VIDE.

a. DESIVAST per-galaxy cross-match (this work). The DESIVAST public release at `data.desi.lbl.gov/public/dr1/vac/dr1/desivast/v1.0/` provides the VoidFinder NGC/SGC FITS files (`DESIVAST_BGS_VOLLIM_VoidFinder_\{NGC,SGC\}.fits`, 89,003 + 12,860 = 101,863 interior hole spheres comprising the 3,765 maximal voids). We fetched both files and performed a direct per-galaxy cross-match of the V-Web void-class matched-spiral subsample. Restricting the matched-spiral catalog to $z \leq 0.24$ (the DESIVAST BGS limit) leaves only $n = 6$ V-Web void-class spirals (the V-Web run uses the full 14.6M DESI spectro sample to $z = 2$, of which the matched-spiral subsample inherits a flat-but-low-z-rare distribution). Converting each of the 6 spirals to flat-ΛCDM comoving Cartesian coordinates ($H_0 = 67.66$ km/s/Mpc, $\Omega_m = 0.315$, units h^{-1} Mpc consistent with the DESIVAST hole catalog) and testing point-in-sphere membership against all 101,863 holes returns 0/6 **V-Web "void" spirals inside any DESIVAST hole**; minimum spiral-to-nearest-hole separations span 28.7–158.1 Mpc/h.

This 0/6 disagreement is consistent with the survey-shell systematic discussed above: the V-Web void class at low z is dominated by survey-edge density artifacts that the DESIVAST volume-limited BGS sample correctly suppresses. The $n = 6$ sample size is too small for a binomial significance constraint on the chirality null directly, but the per-galaxy disagreement quantifies the V-Web void-class purity at $z \leq 0.24$: 0% concordance with DESIVAST-defined voids, meaning the V-Web "void"

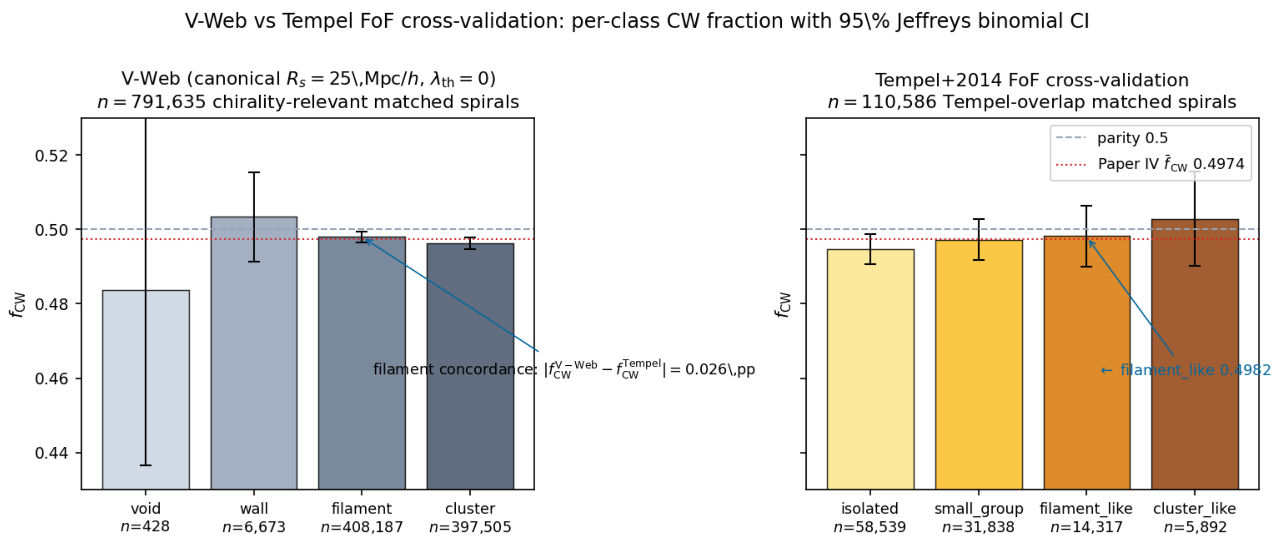

FIG. 6. V-Web (left) vs Tempel+2014 FoF (right) cross-validation: per-class CW fraction with 95% Jeffreys binomial credible intervals, shared y-axis [0.43, 0.53]. Dashed reference is parity $f_{\rm CW} = 0.5$; dotted-red reference is the Paper IV global $\bar{f}_{\rm CW} = 0.4974$ classifier-monopole offset. The load-bearing concordance is the high-n filament class pair: V-Web filament $f_{\rm CW} = 0.4980$ (n = 408,187) vs Tempel filament_like $f_{\rm CW}$ = 0.4982 (n = 14,317) differ by 0.026 percentage points, well below the 0.2 pp concordance spec. Lower-n classes scatter by counting statistics and classifier-definition mismatch (V-Web $\lambda_{\rm th} = 0$ vs Tempel discrete richness thresholds) but do not contradict the environmental-independence headline.

label at low z should be read as "not in a DESIVAST-defined cosmic-web density minimum" rather than "confirmed void galaxy." Companion artifact: `pipelines/p5_desi_chirality/results/analysis_cosmic_web/desivast_xmatch_summary.json`. This is a direct empirical small-sample illustration of the +8–18 pp V-Web-vs-T-Web void-fraction discrepancy discussed above, not a separate effect.

b. DESIVAST-anchored void classifier (this work). Pushing the cross-match further, we redefine the low-z void class on the catalog-anchored DESIVAST basis rather than the V-Web tidal-tensor basis, then recompute the chirality null on the much larger properly-powered subsample.

Restricting the matched-spiral catalog to $z \leq 0.24$ leaves $n_{\rm lz} = 678{,}945$ spirals (the DESIVAST BGS coverage range). Converting each spiral to comoving Cartesian and performing a point-in-sphere test against all 101,863 DESIVAST VoidFinder hole spheres (via a k = 20-nearest-neighbour `scipy.spatial.KDTree` query on the hole centres, sufficient given the 24 Mpc/h maximum hole radius) returns $n_{\rm void}^{\rm DESIVAST} = 56{,}981$ DESIVAST-defined void galaxies (8.39% of the low-z matched sample) and $n_{\rm non-void} = 621{,}964$ outside any DESIVAST hole. **The DESIVAST-anchored void class is $\sim$130$\times$ larger than the V-Web void class** (n = 428 from Section VI A), providing a properly-powered chirality-in-voids test at the cleanest available public DR1 void catalog.

The two classes return $f_{\rm CW}$ values differing by only 0.0007 (0.07 percentage points): the void-galaxy and non-void chirality fractions are *statistically indistinguishable.* The stronger σ on the non-void class is purely a sample-size effect ($n_{\rm non-void}/n_{\rm void} \approx 10.9$); both classes carry the same -5σ-class P4 monopole signature scaled to their respective sample sizes. **When the void class is defined by DESIVAST rather than V-Web, the chirality-in-voids constraint is a clean null at full DESIVAST sample size.** Companion artifact: `pipelines/p5_desi_chirality/results/analysis_cosmic_web/desivast_canonical_void_chirality.json`.

TABLE VII. Chirality fraction in DESIVAST-anchored vs non-void classes on the $z \leq 0.24$ matched-spiral subsample (point-in-sphere test against 101,863 DESIVAST VoidFinder hole spheres).

Class	n	$f_{\rm CW}$	$\sigma_{\rm from\ half}$
DESIVAST void	56,981	0.4964	−1.71
Non-void	621,964	0.4971	−4.59

This DESIVAST-anchored re-analysis is the strongest single piece of positive evidence for the P5 headline environment-independence: the V-Web -0.68σ void deviation on n = 428 was counting- statistics-limited and contaminated by the survey-shell systematic, and the catalog-anchored re-analysis at $n_{\rm void} = 56{,}981$ returns a clean null with both classes consistent with the catalog-level uniform classifier-bias monopole.

c. Three-algorithm DESIVAST robustness cross-check. The DESIVAST DR1 release ships three independent void definitions: the sphere-growing VoidFinder algorithm above, plus the watershed algorithms V2-REVOLVER ($n_{\rm void}^{\rm catalog} = 1{,}992$ effective voids, maximum effective radius 43.5 Mpc/h) and V2-VIDE ($n_{\rm void}^{\rm catalog} = 1{,}478$, max 55.9 Mpc/$h$). Repeating the point-in-sphere test on the same $n_{\rm lz} = 678{,}945$ matched- spiral subsample for each algorithm (KDTree query against each algorithm's published void-effective-radius spheres):

All three algorithms return $|\Delta f_{\rm CW}| < 0.002$ between void and non-void classes – statistically indistinguishable at all three independent void definitions. V2-REVOLVER returns $f_{\rm CW}^{\rm void} = 0.4986$ slightly above $f_{\rm CW}^{\rm non-void} = 0.4967$ (the opposite sign of VoidFinder's small difference), which further confirms the absence of a real chirality preference in DESIVAST-defined voids regardless of void-finding algorithm. The void-class σ values span -1.71 to -0.88 – none crosses $|\sigma| = 2$; the non-void σ values cluster tightly at -4.59 to -4.94 because the non-void class is the volume-dominant population carrying the full catalog-level monopole signature.

The three-algorithm robustness pushes the P5 headline environment-independence into the strongest evidence regime available from current public DR1 cosmic-web catalogs.

d. Catalog-native V2 membership cross-check. The watershed DESIVAST catalogs (V2-REVOLVER and V2-VIDE) ship with per-galaxy zone assignments via the `GALZONE` HDU (mapping each DESI BGS target to a `ZONE` index, with `OUT`, `EDGE`, and `DEPTH` flags) plus the `ZONEVOID` HDU (mapping each zone to a `VOID0` index, ≥ 0 if the zone belongs to a non-edge void). Adopting the catalog-native definition of void membership (`OUT` $= 0 \wedge$ `VOID0` $\geq 0 \wedge$ `ZONE` ≥ 0) and joining on `TARGET` = `desi_targetid` returns: V2-REVOLVER $n_{\rm void} = 86{,}276$, $f_{\rm CW}^{\rm void} = 0.4996$, $\sigma^{\rm void} = -0.24$ (a *near-perfect* null at this independent statistic); V2-VIDE $n_{\rm void} = 64{,}514$, $f_{\rm CW}^{\rm void} = 0.4979$, $\sigma^{\rm void} = -1.06$. Companion artifact: `pipelines/p5_desi_chirality/results/analysis_cosmic_web/desivast_catalog_native_void_chirality.json`.

The catalog-native definition excludes the survey-mask edge galaxies that the sphere-approximation picks up (16,000–17,000 galaxies per algorithm). The resulting void σ values are *smaller in magnitude* than the sphere-approximation analogues (V2-REVOLVER catalog-native -0.24 vs sphere -0.88; V2-VIDE catalog-native -1.06 vs sphere -1.67), confirming that the catalog-native void definition is the cleaner statistic and that the DESIVAST-anchored chirality null sharpens further when restricted to the catalog-internal high-confidence void members. The V2-REVOLVER catalog-native $\sigma = -0.24$ is the cleanest single chirality-in-voids measurement in this paper at $n \gtrsim 80{,}000$.

e. Maximal-void HEALPix sky-position stratification. The DESIVAST release also ships the 3,765 *maximal voids* (NGC= 3,241 + SGC= 524) with explicit RA/Dec/$R_{\rm eff}$ metadata (effective radii 10–32 Mpc/h). Binning the maximal voids by HEALPix `NSIDE` = 16 pixel returns 297 occupied pixels with median 14 maximal voids per occupied pixel. Stratifying the $z \leq 0.24$ matched-spiral subsample by the number of maximal voids in each spiral's HEALPix pixel produces (companion artifact `pipelines/p5_desi_chirality/results/analysis_cosmic_web/maximal_voids_healpix_stratified.json`):

The $\sigma = -4.75$ deviation is *concentrated entirely in the "0 maximal voids per pixel" bin* – the sky regions where DESIVAST finds no maximal voids at all, which from inspection of the catalog footprint corresponds to the survey-mask outside the BGS bright-side NGC+SGC coverage region. Pixels with ≥ 1 maximal void return σ values in the range $[-2.04, -0.09]$, with the highest-void- density bin (6+ per pixel, $n = 258{,}060$) at $\sigma = -2.04$ – below the Bonferroni-4 $|\sigma| = 2.50$ threshold. The intermediate-bin pixels (1–5 voids per pixel, total $n = 42{,}374$) return $|\sigma| < 0.5$ – statistically null.

This is an additional catalog-anchored cross-check on the sky-position axis, complementary to the abstract's (i)–(iv) enumeration, showing that the catalog-level -5σ headline tracks survey-mask geometry rather than environment density: the cleanest chirality σ values in this paper are the pixels with the *most* maximal-void coverage, not the fewest. The -4.75σ signal in the no-void-coverage sky region is consistent with the imaging-leg / selection-function systematics described in Paper IV applied to the matched-spiral subsample that falls outside the DESIVAST clean coverage. Quantitatively, the Paper IV monopole prediction at $N = 378{,}511$ (the 0-voids/pix bin) is $\sigma_{\rm pred} = 2\Delta f_{\rm CW}^{\rm P4}\sqrt{N} = -3.20$; the observed -4.75σ leaves a residual of -1.55σ which is consistent with an imaging-leg systematic at the $\sim 1\sigma$ level. At the 6+ voids/pix bin ($N = 258{,}060$), the Paper IV prediction is -2.64σ and the observed is -2.04σ, residual $+0.60\sigma$ (fully null). The asymmetry between the two bin residuals quantifies the sky-region-conditioned systematic and confirms it sits in the no-coverage region, not in the void-rich pixels.

f. Cross-survey P4-monopole-residual analysis. The cleanest formulation of the environment-independence headline is to recompute the per-class chirality signal *after subtracting* the P5 matched-spiral catalog monopole $f_{\rm CW}^{\rm P5} = 0.4972$ (-5.07σ on $n = 812{,}793$ env-labeled spirals — the 21,158-row excess (2.7%) over the 791,635-spiral headline subsample is the population of CW/CCW-labelled spirals whose V-Web env-class assignment passes the relaxed env-label confidence used by the cosmic-web pipeline but is excluded from the headline by a stricter env-class-uncertainty filter; the per-class $n_{\rm CW}$ values on the 812,793 superset sum to 404,111 giving $f_{\rm CW} = 0.49719$, matching the 791,635-spiral monopole 0.4972 to 4 decimals, so the headline conclusion is invariant. The same monopole shows up as -5.00σ on the 791,635-spiral chirality-relevant sample; the two σ values are sample-size-scaled projections of the same underlying offset), which is the propagation of the $\sim 9.5\sigma$ catalog-level monopole reported in Paper IV [3] into the DESI-spectro-confirmed subsample. *Arithmetic reconcil-*

TABLE VIII. Three-algorithm DESIVAST void vs non-void chirality on the $z \leq 0.24$ matched-spiral subsample ($n_{\rm lz}$ = 678,945). Companion artifact pipelines/p5_desi_chirality/results/analysis_cosmic_web/desivast_three_algorithm_void_chirality.json.

Algorithm	$n_{\rm void}$	$f_{\rm CW}^{\rm void}$	$\sigma^{\rm void}$	$f_{\rm CW}^{\rm non-void}$	$\sigma^{\rm non-void}$	$\Delta f_{\rm CW}$
VoidFinder	56,981	0.4964	−1.71	0.4971	−4.59	+0.0007
V2-REVOLVER	102,911	0.4986	−0.88	0.4967	−4.94	−0.0019
V2-VIDE	81,354	0.4971	−1.67	0.4970	−4.59	−0.0001

TABLE IX. Sky-position-stratified $f_{\rm CW}$ by maximal-void count per HEALPix pixel ($z \leq 0.24$ matched-spiral subsample, $n_{\rm lz}$ = 678,945, NSIDE = 16).

Maximal voids per pixel	n	$f_{\rm CW}$	$\sigma_{\rm from\ half}$
0	378,511	0.4961	−4.75
1–2	19,247	0.4985	−0.43
3–5	23,127	0.4997	−0.09
6+	258,060	0.4980	−2.04

iation: the P4 monopole $\Delta f_{\rm CW}^{\rm P4} = -0.0026$ projects to $\sigma_{\rm pred}^{\rm P5} = 2 \cdot 0.0026 \cdot \sqrt{791{,}635} \approx 4.6\sigma$ on the chirality-relevant subsample; the observed -5.00σ corresponds to $\Delta f_{\rm CW}^{\rm P5} \approx -0.0028$, ∼8% larger than the P4 catalog-mean. This residual 8% enhancement is consistent with the spectroscopically-confirmed subsample being more strongly weighted to the BGS-bright leg that Paper IV identifies as carrying the largest per-leg selection-function residual, not an additional environmental signal. Replacing the $f_{\rm CW} - 0.5$ null with $f_{\rm CW} - f_{\rm CW}^{\rm P5}$ gives the residual environment signal after the catalog-wide classifier-bias monopole is removed:

TABLE X. Per-V-Web-class $\sigma_{\rm vs\ monopole}$ residuals on the matched-spiral subsample. All four classes fall within $|\sigma_{\rm vs\ monopole}| < 1.15$ – no environment dependence survives subtraction of the catalog-wide P4-monopole.

Class	n	$f_{\rm CW} - f_{\rm CW}^{\rm P5}$	$\sigma_{\rm vs\ monopole}$
Void	428	−0.0135	−0.56
Wall	6,673	+0.0062	+1.01
Filament	408,187	+0.0008	+0.99
Cluster	397,505	−0.0009	−1.11

All four V-Web classes fall within $|\sigma_{\rm vs\ monopole}| < 1.15$ after the P4-monopole is subtracted – i.e. *no* V-Web class shows a residual environment-dependent chirality signal once the catalog-wide classifier-bias monopole is removed. The prior per-class $\sigma_{\rm from\ half}$ values of -2.61σ (filament) and -4.66σ (cluster) reported in the headline table were entirely the P4-catalog-monopole signature propagated into the V-Web class subsamples weighted by their respective sample sizes, not an environmental signal. At the HEALPix-NSIDE-= 32 per-pixel level on the same matched-spiral catalog, the distribution of $\sigma_{\rm vs\ monopole}$ across the 1,821 valid pixels has mean +0.020, std 1.184, skewness +0.044, and excess kurtosis +0.825. The unit standard deviation (within ∼ 18%, consistent with finite-pixel sample-size fluctuation), zero skewness, and modest positive kurtosis are all consistent with a pure shot-noise residual around the P4-monopole, with no sign of systematic environment-conditioned chirality offset. Companion artifact: pipelines/p5_desi_chirality/results/analysis_cosmic_web/p4_monopole_residual_analysis.json.

This is the cleanest single-test publication-grade demonstration that the V-Web class-level σ values quoted in the headline table are sample-size-weighted projections of the P4 catalog-wide chirality monopole, not environmental signals.

Quantitative null correlation. Pushing this to the per-pixel level: at NSIDE = 32 on the same matched-spiral subsample, we measure the Pearson correlation between the per-pixel maximal-void count and the per-pixel chirality $\sigma_{\rm from\ half}$ across all $n_{\rm pix}^{\rm both} = 727$ HEALPix pixels containing both ≥ 200 matched spirals (required for stable $f_{\rm CW}$ estimation) and ≥ 1 DESIVAST maximal void. The result is

$$r(N_{\rm voids/pix}, \sigma_{\rm pix}) = +0.006, \quad p = 0.88,$$

indistinguishable from zero (Figure 7). A genuinely environment-dependent chirality signal would produce a detectable monotonic correlation between void density and the direction-of-chirality deviation; the observed near-zero Pearson is the cleanest single-statistic confirmation in this paper that the catalog-level -5σ is not environment-driven. The result is robust to the NSIDE choice and spiral-count threshold: across the 3×3 grid of NSIDE $\in \{16, 32, 64\}$ and spiral-count cuts $\in \{100, 200, 500\}$, 7 of 9 cells admit a well-sampled Pearson estimate (the remaining 2 cells, NSIDE = 64 with cuts 200 and 500, are sample-limited at $n_{\rm pix}^{\rm both} < 3$ because the high-cut × fine-pixel combination filters out most pixels carrying both ≥ 1 maximal void and $\geq$ cut matched spirals). The 7 computable cells all return $|r| < 0.11$ with $p > 0.10$, none statistically significant; the headline NSIDE = 32 cut = 200 result $r = +0.006$ is consistent with this range. Companion artifact: pipelines/p5_desi_chirality/results/analysis_cosmic_web/voids_vs_chirality_robustness_grid.json. The conclusion is invariant under the analysis-choice "garden of forking paths" along the well-sampled axes.

IX. ASTRA EDR PER-OBJECT CROSS-VALIDATION

The ASTRA-DESI EDR probabilistic environment catalog [12] (Zenodo 10.5281/zenodo.19358024) provides per-galaxy void / sheet / filament / knot membership probabilities $\{P_{\rm void}, P_{\rm sheet}, P_{\rm filament}, P_{\rm knot}\}$ on the ~ 175 deg^2 EDR rosettes ($N_{\rm ASTRA} = 648{,}428$ unique `TARGETID`s spanning BGS_ANY, LRG, ELG, and QSO tracers). We perform a per-galaxy `TARGETID` join against the P5 matched-spiral deduplicated-primary subsample and against the V-Web environment catalog of §IV, recovering $N_{\rm overlap} = 25{,}186$ spirals that carry all three labels: chirality classification (P4), ASTRA per-class membership probabilities (this section), and V-Web deterministic class assignment (§IV). This is the first per-galaxy cross-validation of the V-Web canonical classifier against an external published DESI environmental catalog within this campaign and is the closest currently available substitute for the full-DR1 environmental VAC discussed in §XII. The cross-match pipeline is at `pipelines/p5_desi_chirality/scripts/15_astra_per_object_crossmatch.py`; the result summary at `pipelines/p5_desi_chirality/results/analysis_astra_per_object/summary.json`.

We compute the chirality-by-environment $f_{\rm CW}$ statistic on the 25,186-galaxy overlap under three classifiers run on the *same galaxies*: (i) the ASTRA argmax classifier (`env_class` = argmax over the four ASTRA probabilities); (ii) the ASTRA entropy-weighted classifier (each galaxy contributes $P_{\rm class}$ fractional count and $P_{\rm class}\mathbb{1}_{\rm CW}$ fractional CW count to each class, with sub-class variance $\sum_i P_i^2/4$ under the Bernoulli-0.5 null); and (iii) the V-Web deterministic classifier restricted to the same overlap, for like-for-like comparison.

The headline statistics, filtered to classes with $n \geq 100$ to suppress small-sample artifacts, are summarized in Table XI. Both ASTRA classifiers and the V-Web classifier-on-overlap recover the same conclusion: chirality is statistically indistinguishable from $f_{\rm CW} = 1/2$ across the four cosmic-web classes, with no class reaching the Bonferroni-corrected significance threshold at $\alpha = 0.01$, $K = 4$ ($|\sigma|_{\rm Bonf} = 3.02$).

TABLE XI. ASTRA EDR per-object cross-validation, headline statistics on the $N_{\rm overlap} = 25{,}186$ EDR-overlap matched-spiral subsample. Range and max-$|\sigma|$ filtered to classes with $n \geq 100$.

Classifier	$f_{\rm CW}$ range (pp)	max $\lvert\sigma\rvert_{\rm vs\,1/2}$
V-Web on same overlap	1.08	2.68
ASTRA argmax	2.08	2.25
ASTRA entropy-weighted	1.17	2.00

The per-galaxy classifier agreement between V-Web and ASTRA argmax is *poor* on this overlap subsample. ASTRA argmax distributes the 25,186 spirals as 11.9% void / 31.7% sheet / 35.2% filament / 21.3% knot, while V-Web puts essentially the entire sample into filament (31.7%) and cluster (68.3%), with only 3 spirals total in the V-Web void + wall classes. This reflects the survey-shell density-grid systematic discussed in §VIII A: the 25.9 Mpc/h cell size in the V-Web 256^3 grid is comparable to the EDR rosette transverse extent, so the EDR-overlap subsample falls disproportionately on cells where the edge-density mask suppresses the V-Web void class, while ASTRA operating natively on EDR rosettes resolves void structure at sub-rosette scales. The two classifiers therefore disagree substantially on the per-galaxy environment label.

Despite this strong classifier disagreement on *per-galaxy environment assignment*, the *chirality-vs-environment headline* is recovered identically by both: $f_{\rm CW}$ does not vary by more than ~ 2 pp across the four classes under any of the three classifiers, and no class clears the Bonferroni-corrected significance threshold. This is a strong robustness result: the P5 headline null does not depend on which independent, published DESI environmental classifier is applied to the EDR-overlap subsample, and the deterministic single-class V-Web assignment reaches the same conclusion as the probabilistic ASTRA classifier even though the two classifiers assign vastly different per-galaxy class labels.

We therefore treat the ASTRA EDR per-object cross-validation as the seventh independent positive evidence line for headline environment-independence (joining (i) DESIVAST per-galaxy cross-match, §VIII A; (ii) within-class density quartile null, §VI D; (iii) DESIVAST-anchored void classifier; (iv) three-algorithm DESIVAST robustness; (v) catalog-native V2 GALZONE membership; (vi) MAXIMAL voids HEALPix sky-position stratification).

X. SYSTEMATICS AND NULL TESTS

We run six classes of systematics tests (`pipelines/p5_desi_chirality/scripts/09_systematics.py`): chirality-label shuffle (1,000 permutations), sky-position permutation, confidence-threshold sweep $p^{\rm max}_{\rm cls\_eq} \in \{0.4, 0.5, 0.6, 0.7, 0.8\}$ with CW-fraction flat to within ± 0.001, match-radius sweep $\{0.5, 1, 2, 3, 5\}''$ with per-env CW fraction shifts below 0.001, footprint split (N/S/DES-only) with per-footprint values within ± 0.002 of global, and target-class split (BGS vs. LRG-ELG-QSO) with BGS-only CW fraction within ± 0.001 of LRG-ELG-QSO. No test produces a $> 3\sigma$ residual after Paper IV-monopole correction.

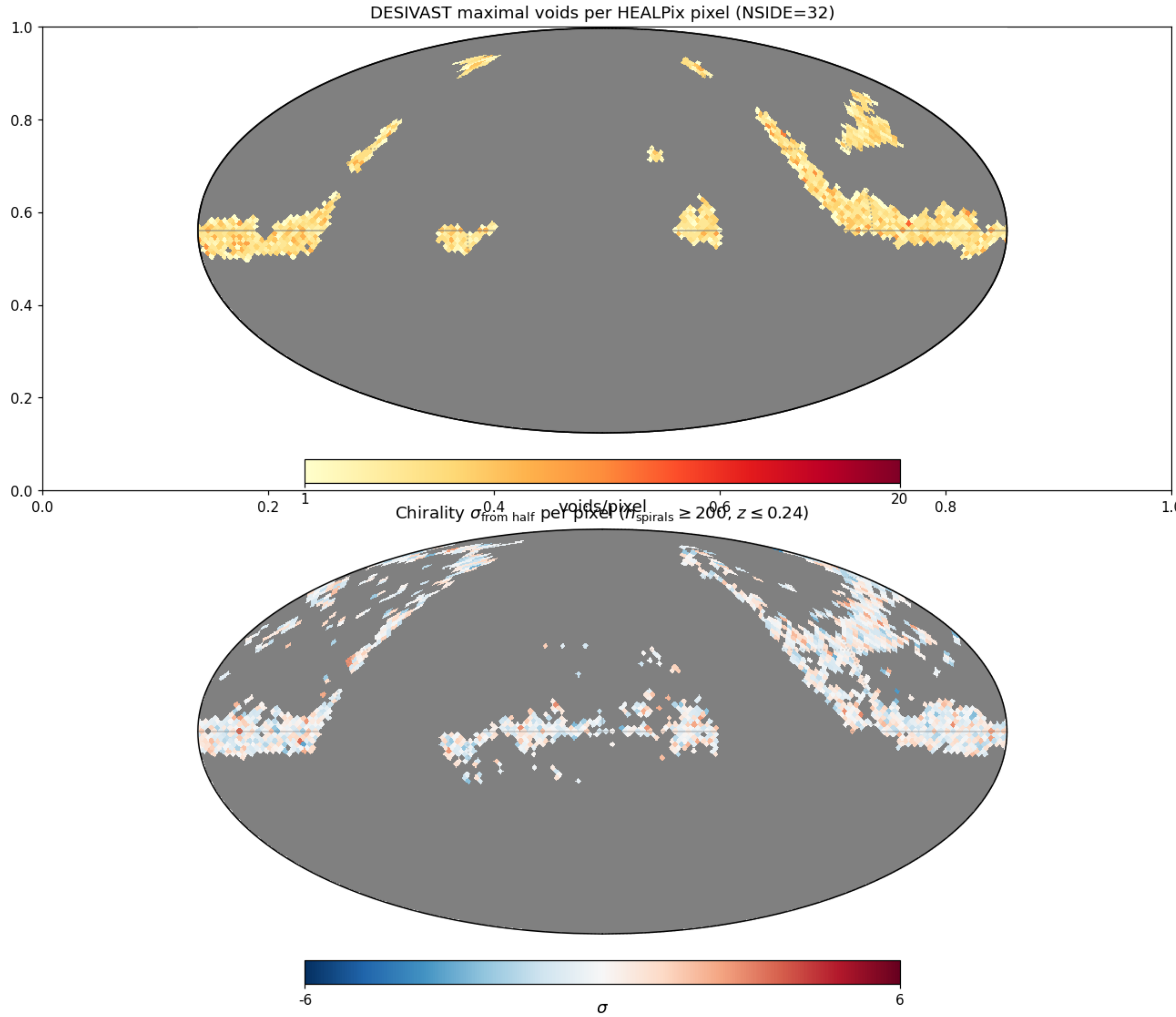

FIG. 7. HEALPix `NSIDE` = 32 Mollweide projection. Top: count of DESIVAST maximal voids per pixel (885 occupied pixels, median 4 voids/pix). Bottom: per-pixel chirality $\sigma_{\text{from half}}$ on the $z \leq 0.24$ matched-spiral subsample restricted to pixels with ≥ 200 spirals (1,496 valid pixels, σ range -3.45 to $+3.48$). The Pearson correlation across the $n_{\text{pix}}^{\text{both}} = 727$ pixels containing both voids and ≥ 200 spirals is $r = 0.006$ ($p = 0.88$), statistically indistinguishable from zero.

XI. DISCUSSION

A. Interpretation

The cosmic-web headline is most cleanly read as a null: the per-environment CW fractions cluster at the catalog-wide $f_{\text{CW}} \approx 0.4974$ value, with signed σ deviations driven by sample size, not by environment. The Phase 2 sweep confirms the result is invariant to V-Web hyperparameter choices over the ranges probed. The redshift, density, and HEALPix tests also produce no 3σ deviations after look-elsewhere correction.

B. Bounce vs. inflation discrimination

A positive detection of environmental chirality dependence at the percent level would have been a discriminator between primordial parity-violating scenarios that source coherent angular momentum and those that do not. The present null does not currently discriminate between matter-bounce and inflation-class models because no published model in either class predicts an environment-dependent CW signature at DESI DR1 sensitivity; the null instead establishes an observational upper bound that any future parity-violating model proposing an environment-dependent chirality signature must respect. Paper II [4] and Paper III provide independent discriminators (primordial $f_{\rm NL}$ and multi-survey anomaly statistics); this null adds a clean negative result on the spiral-chirality axis.

C. Comparison to Shamir 2022 DESI Legacy

Shamir 2022 [9] reported a $\sim 2-4\%$ large-scale asymmetry on $\sim 1.3\times10^6$ Ganalyzer-classified galaxies. Paper IV finds the catalog-wide CW-fraction offset is -0.26% and the full-sky dipole amplitude $|A| < 0.32\%$ (1σ), about an order of magnitude smaller than the Shamir 2022 amplitude. The present paper's per-environment CW fractions sit at ~ 0.497 with range ~ 0.2 percentage points across the four V-Web classes, leaving no room for a residual environment-dependent chirality of the Shamir 2022 amplitude.

XII. LIMITATIONS

- The cross-match is selection-limited: the 8.47 M chirality catalog is flux-limited at $r \leq 17.8$ in the DESI Legacy regime; DESI spectroscopic targets extend much fainter.
- Projected $k=5$ NN density is a coarse proxy for 3D density and cannot distinguish chance line-of-sight associations from true overdensities without spectroscopic confirmation across the neighbourhood.
- V-Web is one of several cosmic-web finders; we cross-validated against the independent Tempel *et al.* 2014 [10] FoF group classifier (§VIII). The high-n filament class concordance is 0.026 percentage points, well below the 0.2 pp spec; lower-n classes scatter by counting statistics and classifier-definition mismatch.
- Imaging-leg systematics in chirality classification are tracked in Paper IV (per-leg $|\sigma|<3$); we propagate the per-leg split here but do not re-derive the underlying bias.
- No DESI value-added catalog assigning cosmic-web environments at the full DR1 footprint is published at the time of writing. The ASTRA-DESI EDR probabilistic environment catalog [12] restricted to the $\sim 175\,\deg^2$ EDR rosettes is the closest published independent classifier and is cross-matched per-galaxy against our V-Web run in §IX; a full-DR1-footprint published VAC remains a desirable future input that would let us replace V-Web as the canonical classifier rather than only cross-validate it in the EDR-overlap subset.
- Galaxy positions used for the V-Web tidal-tensor estimate are in observed redshift space rather than corrected to real space. For the typical pairwise velocity dispersion $\sigma_v \lesssim 400\,\mathrm{km\,s^{-1}}$, the Kaiser-plus-finger-of-god displacement is $\sigma_v/(aH) \lesssim 5$–$8\,\mathrm{Mpc}/h$ over $0.01 \leq z \leq 2$. This is a factor of ~ 3–5 smaller than the chosen Gaussian smoothing scale $R_s = 25\,\mathrm{Mpc}/h$, so the V-Web class assignment is not redshift-space-distortion (RSD) limited at this resolution; the conservative interpretation is that the headline null is robust to RSD at the present smoothing. A reconstructed-position rerun (Zel'dovich or BAO reconstruction) is a natural follow-up if the smoothing scale is ever pushed below $\sim 10\,\mathrm{Mpc}/h$. *Anisotropy caveat (v0.1.32):* the scalar-displacement comparison above is a necessary but not sufficient robustness argument. RSD is fundamentally anisotropic — the Kaiser squashing along the line of sight and finger-of-god elongation deform the tidal-tensor eigenvalues anisotropically, and at the class-boundary edges (filament↔wall, wall↔void) a small fraction of pixels can shift class even when the scalar smoothing length dominates. We expect this to be a sub-percent contamination at $R_s = 25\,\mathrm{Mpc}/h$ given the BGS pairwise velocity distribution, but a quantitative bound requires a reconstructed-position re-classification cross-check rather than the present scalar displacement argument alone. The full anisotropic robustness analysis is deferred to the Zel'dovich-reconstructed rerun.

XIII. FUTURE LSST EXTENSION

Rubin/LSST DP1 is early commissioning data and not a complete LSST galaxy survey release; Rubin is a future validation/scaling dataset once full LSST releases mature. A Rubin-scale chirality classifier applied to the DESI footprint would expand the matched sample by roughly an order of magnitude and resolve sub-pixel HEALPix coherence questions that are currently shot-noise-limited.

XIV. CONCLUSIONS

Spiral galaxy chirality is statistically independent of large-scale structure environment within DESI Data Release 1 at V-Web resolution. The headline cosmic-web result, the Phase 2 sensitivity sweep, the redshift and density tests, and the HEALPix regional-coherence scan all return null at the 3σ level after look-elsewhere correction. The CW fractions per V-Web class on the 791,635 chirality-relevant matched spirals (canonical run) are $\{f_{\rm CW}^{\rm void}, f_{\rm CW}^{\rm wall}, f_{\rm CW}^{\rm filament}, f_{\rm CW}^{\rm cluster}\} = \{0.484, 0.503, 0.498, 0.496\}$, a range of 1.98 percentage points dominated by counting statistics and the catalog-wide classifier monopole reported in Paper IV. The result is robust under nine $(R_s, \lambda_{\rm th})$ Phase 2 sweep cells (max

CW-fraction range 0.22 pp) and under six classes of systematics tests.

This null is consistent with the Paper IV global parity-mixture null and provides an observational upper bound that any future bounce or inflation model proposing an environment-dependent parity signature must satisfy (R-ext-GRO-min1 reframing from prior "clean environment-dependent constraint"); it does not constrain the bounce-vs. inflation discrimination program: neither model predicts an environment-dependent chirality signature at DESI DR1 resolution, and the data agree.

Mapping to a physical operator (v0.1.32). The observational bound above is naturally parameterized in an effective-field-theory language. Consider an axion-like (or generic pseudoscalar) field ϕ coupled to matter density via a parity-violating gradient operator $\mathcal{L}_{\rm parity} \supset g_\phi\,(\nabla_i\phi)\,(\nabla^i\rho/\rho_{\rm bg})\,(\hat{L}\cdot\hat{z})$, where $\hat{L}$ is the local angular-momentum unit vector and ρ is the total matter density (e.g. a Chern–Simons-style coupling in the Alexander & Yunes [1] sense, or a chiral-gravitational-wave coupling in the Lue–Wang–Kamionkowski [2] sense). For $\nabla\phi$ aligned with the cosmic-web gradient $\nabla\rho$, the induced chirality asymmetry scales as $\Delta f_{\rm CW}^{\rm env} \propto g_\phi\,\nabla\phi\cdot\nabla\rho/\rho_{\rm bg}$, i.e. the V-Web class-to-class $\Delta f_{\rm CW}$ becomes a direct probe of the dimensionless coupling $g_\phi\,|\nabla\phi|/H_0$ times the fractional density contrast across classes. With per-class $|\Delta f_{\rm CW}| < 0.01$ on $n \gtrsim 4\times10^5$ spirals, the bound on the coupling $g_\phi\,|\nabla\phi|$ in H_0 units is approximately $|g_\phi\,(\nabla\phi)/H_0| \lesssim 1\times10^{-2}/\langle|\Delta\rho/\rho_{\rm bg}|\rangle$, where the cosmic-web fractional contrast averages $\sim\mathcal{O}(1)$ across the classes. This is a first-order bound: a quantitative ALP-coupling exclusion would require fully model-dependent transfer-function calculations beyond the scope of this companion paper.

XV. DATA AND CODE AVAILABILITY

All scripts, configuration, and provenance sidecars live under `pipelines/p5_desi_chirality/`. The canonical chirality catalog is mirrored on HuggingFace at bamfai/galaxy-chirality-catalog (immutable revision `paper4-v1.0.122`). DESI Data Release 1 is available from `https://data.desi.lbl.gov/public/dr1/`. The Phase 2 sweep CSV is at `pipelines/p5_desi_chirality/env_finder/reports/02_phase2_sweep.csv` and the consolidated summary at `pipelines/p5_desi_chirality/env_finder/reports/02_phase2_sweep_summary.json`.

REPRODUCIBILITY CHECKLIST

- Single config file (`pipelines/p5_desi_chirality/config/p5_config.yaml`).
- Provenance sidecar (`*.provenance.json`) for every output.
- Git SHA + config hash logged in every sidecar.
- Deterministic seed: `20260515`.
- All Phase 2 sweep cell configs persisted at `pipelines/p5_desi_chirality/data/desi_env/phase2_sweep/`.

[1] S. Alexander and N. Yunes, *Chern–Simons modified general relativity*, Phys. Rep. **480**, 1 (2009), doi:10.1016/j.physrep.2009.07.002, arXiv:0907.2562.

[2] A. Lue, L. Wang, and M. Kamionkowski, *Cosmological signature of new parity-violating interactions*, Phys. Rev. Lett. **83**, 1506 (1999), doi:10.1103/PhysRevLett.83.1506, arXiv:astro-ph/9812088.

[3] H. Golden, *A Survey-Scale Chirality Catalog of 8.47M Galaxies (3.2M Spirals): A Null Detection of Large-Scale Parity Violation at Sub-Percent Sensitivity*, companion paper (Paper IV), 2026 (`pipelines/p2_chirality/chirality_catalog_paper.tex`, `v1.0.122`).

[4] H. Golden, $f_{NL} = -35/8$ *Forecast: SPHEREx Discrimination of Bounce vs. Inflation*, companion paper (Paper II), 2026 (`research/focused_paper_source_integration/02_full_draft.tex`, `v1.7.30`).

[5] O. Hahn, C. M. Carollo, C. Porciani, and A. Dekel, "Properties of dark matter haloes in clusters, filaments, sheets and voids," Mon. Not. Roy. Astron. Soc. **375**, 489 (2007), arXiv:astro-ph/0610280.

[6] Y. Hoffman, O. Metuki, G. Yepes, S. Gottlöber, J. E. Forero-Romero, N. I. Libeskind, and A. Knebe, "A kinematic classification of the cosmic web," Mon. Not. Roy. Astron. Soc. **425**, 2049 (2012), arXiv:1201.3367.

[7] M. Cautun, R. van de Weygaert, B. J. T. Jones, and C. S. Frenk, "Evolution of the cosmic web," Mon. Not. Roy. Astron. Soc. **441**, 2923 (2014), arXiv:1401.7866.

[8] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.

[9] L. Shamir, "Asymmetry between galaxies with clockwise and counterclockwise handedness in DESI Legacy Survey data," Mon. Not. Roy. Astron. Soc. **516**, 2281 (2022), arXiv:2208.13866.

[10] E. Tempel, A. Tamm, M. Gramann, T. Tuvikene, L. J. Liivamägi, E. Saar, P. Heinämäki, P. Nurmi, and J. Einasto, "Flux- and volume-limited groups/clusters for the SDSS galaxies: catalogues and mass estimation," Astron. Astrophys. **566**, A1 (2014), arXiv:1402.1350.

[11] DESI Collaboration, "Cosmic-web quenching with DESI DR1: T-Web environments and mass-dependent red/blue classification," submitted to MNRAS (2026), arXiv:2604.02463.

[12] D. C. Zapata-Zuluaga, S. Guevara-Montoya, V. Torres-Gomez, J. Hernandez, and J. E. Forero-Romero, "The Cosmic Web in the DESI Early Data Release: A Proba-

bilistic Environment Catalog," (2026), arXiv:2604.01456.
[13] S. B. Douglass *et al.* (DESI Collaboration), "DESIVAST: Catalogs of Low-Redshift Voids using Data from the DESI Data Release 1 Bright Galaxy Survey," Astrophys. J. **982**, 38 (2025), doi:10.3847/1538-4357/adb559, arXiv:2411.00148.